

Chapter 1

Basics of Networking

Learning Outcomes

After reading this chapter, the reader will be able to:

- Understand the basic principles of computer networking
- List the basic terminologies and technologies
- Relate new concepts of IoT with the basics of networking
- Discuss various network configurations and topologies
- Explain various OSI (open systems interconnections) and TCP/IP (transmission control protocol/Internet protocol) layers and their associated uses
- Describe basics of network addressing

1.1 Introduction

In the present era of data- and information-centric operations, everything—right from agriculture to military operations—relies heavily on information. The quality of any particular information is as good as the variety and strength of the data that generates this information. Additionally, the speed at which data is updated to all members of a team (which may be a group of individuals, an organization, or a country) dictates the advantage that the team has over others in generating useful information from the gathered data. Considering the present-day global scale of operations of various organizations or militaries of various countries, the speed and nature of germane information are crucial for maintaining an edge over others in the same area. To sum it up, today's world relies heavily on data and networking, which allows for the instant availability of information from anywhere on the earth at any moment.

Typically, networking refers to the linking of computers and communication network devices (also referred to as hosts), which interconnect through a network

(Internet or Intranet) and are separated by unique device identifiers (Internet protocol, IP addresses and media access control, MAC addresses). These hosts may be connected by a single path or through multiple paths for sending and receiving data. The data transferred between the hosts may be text, images, or videos, which are typically in the form of binary bit streams [1].

Points to ponder

- The data generated from a camera sensor tells us more about a scene compared to the data generated from, say, a proximity sensor, which only detects the presence of people in its sensing range.
- Furthermore, the simultaneous data generated from multiple cameras focusing on the same spot from various angles tell us even more about the scene than a single camera focused at that scene.

As the primary aim of this chapter is to provide the reader with an overview of networking, we have structured the text in such a manner that the general concepts are covered. Additional *Check yourself* suggestions to review various associated technologies are provided along with the topics.

We start our discussion with the different types of networks, followed by an overview of two popularly used layered network models: ISO-OSI (the open systems interconnection developed by the International Organization of Standardization) and TCP/IP (transmission control protocol/Internet protocol) suite. Subsequently, we will touch upon the various types of addressing mechanisms and set up the basic premise of how a message is transmitted between two devices/computers/hosts.

1.2 Network Types

Computer networks are classified according to various parameters: 1) Type of connection, 2) physical topology, and 3) reach of the network. These classifications are helpful in deciding the requirements of a network setup and provide insights into the appropriate selection of a network type for the setup.

1.2.1 Connection types

Depending on the way a host communicates with other hosts, computer networks are of two types—(Figure 1.1): *Point-to-point* and *Point-to-multipoint*.

- (i) **Point-to-point:** Point-to-point connections are used to establish direct connections between two hosts. Day-to-day systems such as a remote control for an air conditioner or television is a point to point connection, where the connection has the whole channel dedicated to it only. These networks were

designed to work over duplex links and are functional for both synchronous as well as asynchronous systems. Regarding computer networks, point to point connections find usage for specific purposes such as in optical networks.

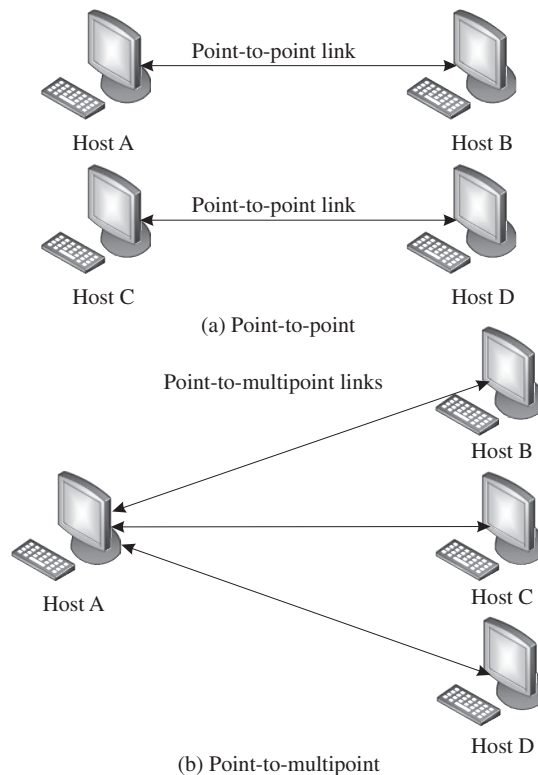


Figure 1.1 Network types based on connection types

Point-to-point Requests for Comments (RFCs)

The following requests for comments (RFCs) are associated with point-to-point communication and its derivatives. **RFC 1332**: point-to-point (PPP) Internet protocol control protocol (IPCP); **RFC 1661**: PPP; **RFC 5072**: IP Version 6 over PPP; **RFC 2516**: PPP over Ethernet; **RFC 1963**: PPP serial data transport protocol; **RFC 1962**: PPP compression control protocol (CCP); **RFC 1990**: PPP multilink protocol (MP); **RFC 2615**: PPP over SONET/SDH (synchronous optical networking/synchronous digital hierarchy).

- (ii) **Point-to-multipoint**: In a point-to-multipoint connection, more than two hosts share the same link. This type of configuration is similar to the one-to-many connection type. Point-to-multipoint connections find popular use in wireless networks and IP telephony. The channel is shared between the various hosts,

either spatially or temporally. One common scheme of spatial sharing of the channel is frequency division multiple access (FDMA). Temporal sharing of channels include approaches such as time division multiple access (TDMA). Each of the spectral and temporal sharing approaches has various schemes and protocols for channel sharing in point-to-multipoint networks. Point-to-multipoint connections find popular use in present-day networks, especially while enabling communication between a massive number of connected devices.

Check yourself

Space division multiplexing, Frequency division multiplexing, Time division multiplexing, Polarization division multiplexing, Orbital angular momentum multiplexing, Code division multiplexing

1.2.2 Physical topology

Depending on the physical manner in which communication paths between the hosts are connected, computer networks can have the following four broad topologies—(Figure 1.2): *Star*, *Mesh*, *Bus*, and *Ring*.

- (i) **Star:** In a star topology, every host has a point-to-point link to a central controller or hub. The hosts cannot communicate with one another directly; they can only do so through the central hub. The hub acts as the network traffic exchange. For large-scale systems, the hub, essentially, has to be a powerful server to handle all the simultaneous traffic flowing through it. However, as there are fewer links (only one link per host), this topology is cheaper and easier to set up. The main advantages of the star topology are easy installation and the ease of fault identification within the network. If the central hub remains uncompromised, link failures between a host and the hub do not have a big effect on the network, except for the host that is affected. However, the main disadvantage of this topology is the danger of a single point of failure. If the hub fails, the whole network fails.
- (ii) **Mesh:** In a mesh topology, every host is connected to every other host using a dedicated link (in a point-to-point manner). This implies that for n hosts in a mesh, there are a total of $n(n-1)/2$ dedicated full duplex links between the hosts. This massive number of links makes the mesh topology expensive. However, it offers certain specific advantages over other topologies. The first significant advantage is the robustness and resilience of the system. Even if a link is down or broken, the network is still fully functional as there remain other pathways for the traffic to flow through. The second advantage is the security and privacy of the traffic as the data is only seen by the intended recipients and not by all members of the network. The third advantage is the reduced data load on a

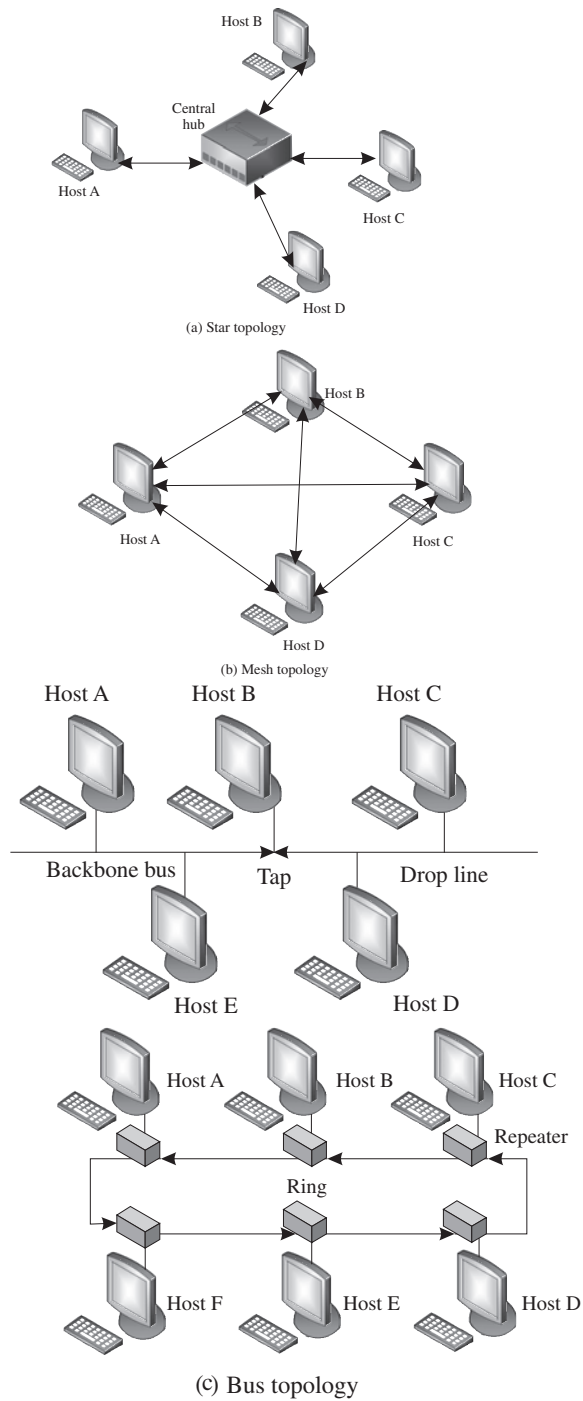


Figure 1.2 Network types based on physical topologies

single host, as every host in this network takes care of its traffic load. However, owing to the complexities in forming physical connections between devices and the cost of establishing these links, mesh networks are used very selectively, such as in backbone networks.

- (iii) **Bus:** A bus topology follows the point-to-multipoint connection. A backbone cable or bus serves as the primary traffic pathway between the hosts. The hosts are connected to the main bus employing drop lines or taps. The main advantage of this topology is the ease of installation. However, there is a restriction on the length of the bus and the number of hosts that can be simultaneously connected to the bus due to signal loss over the extended bus. The bus topology has a simple cabling procedure in which a single bus (backbone cable) can be used for an organization. Multiple drop lines and taps can be used to connect various hosts to the bus, making installation very easy and cheap. However, the main drawback of this topology is the difficulty in fault localization within the network.
- (iv) **Ring:** A ring topology works on the principle of a point-to-point connection. Here, each host is configured to have a dedicated point-to-point connection with its two immediate neighboring hosts on either side of it through repeaters at each host. The repetition of this system forms a ring. The repeaters at each host capture the incoming signal intended for other hosts, regenerates the bit stream, and passes it onto the next repeater. Fault identification and set up of the ring topology is quite simple and straightforward. However, the main disadvantage of this system is the high probability of a single point of failure. If even one repeater fails, the whole network goes down.

Table 1.1 compares the various network topologies.

Table 1.1 Network topology comparison

Topology	Feature	Advantage	Disadvantage
Star	Point-to-point	Cheap; ease of installation; ease of fault identification	Single point of failure; traffic visible to network entities
Mesh	Point-to-point	Resilient against single point of failures; scalable; traffic privacy and security ensured	Costly; complex connections
Bus	Point-to-multipoint	Ease of installation; cheap	Length of backbone cable limited; number of hosts limited; hard to localize faults
Ring	Point-to-point	Ease of installation; cheap; ease of fault identification	Prone to single point of failure

1.2.3 Network reachability

Computer networks are divided into four broad categories based on network reachability: *personal area networks*, *local area networks*, *wide area networks*, and *metropolitan area networks*.

- (i) **Personal Area Networks (PAN):** PANs, as the name suggests, are mostly restricted to individual usage. A good example of PANs may be connected wireless headphones, wireless speakers, laptops, smartphones, wireless keyboards, wireless mouse, and printers within a house. Generally, PANs are wireless networks, which make use of low-range and low-power technologies such as Bluetooth. The reachability of PANs lies in the range of a few centimeters to a few meters.
- (ii) **Local Area Networks (LAN):** A LAN is a collection of hosts linked to a single network through wired or wireless connections. However, LANs are restricted to buildings, organizations, or campuses. Typically, a few leased lines connected to the Internet provide web access to the whole organization or a campus; the lines are further redistributed to multiple hosts within the LAN enabling hosts. The hosts are much more in number than the actual direct lines to the Internet to access the web from within the organization. This also allows the organization to define various access control policies for web access within its hierarchy. Typically, the present-day data access rates within the LANs range from 100 Mbps to 1000 Mbps, with very high fault-tolerance levels. Commonly used network components in a LAN are servers, hubs, routers, switches, terminals, and computers.
- (iii) **Metropolitan Area Networks (MAN):** The reachability of a MAN lies between that of a LAN and a WAN. Typically, MANs connect various organizations or buildings within a given geographic location or city. An excellent example of a MAN is an Internet service provider (ISP) supplying Internet connectivity to various organizations within a city. As MANs are costly, they may not be owned by individuals or even single organizations. Typical networking devices/components in MANs are modems and cables. MANs tend to have moderate fault tolerance levels.
- (iv) **Wide Area Networks (WAN):** WANs typically connect diverse geographic locations. However, they are restricted within the boundaries of a state or country. The data rate of WANs is in the order of a fraction of LAN's data rate. Typically, WANs connecting two LANs or MANs may use public switched telephone networks (PSTNs) or satellite-based links. Due to the long transmission ranges, WANs tend to have more errors and noise during transmission and are very costly to maintain. The fault tolerance of WANs are also generally low.

Check yourself

ARPANET, BITNET, Cellular network, CYCLADES, FidoNet, Telex, World Wide Web

1.3 Layered Network Models

The intercommunication between hosts in any computer network, be it a large-scale or a small-scale one, is built upon the premise of various task-specific layers. Two of the most commonly accepted and used traditional layered network models are the open systems interconnection developed by the International Organization of Standardization (ISO-OSI) reference model and the Internet protocol suite.

1.3.1 OSI Model

The ISO-OSI model is a conceptual framework that partitions any networked communication device into seven layers of abstraction, each performing distinct tasks based on the underlying technology and internal structure of the hosts. These seven layers, from bottom-up, are as follows: 1) *Physical layer*, 2) *Data link layer*, 3) *Network layer*, 4) *Transport layer*, 5) *Session layer*, 6) *Presentation layer*, and 7) *Application layer*. The major highlights of each of these layers are explained in this section.

Points to ponder

The OSI or open system interconnect model for networked devices was standardized by the International Standards Organization (ISO). It is a conceptual framework that divides any networked communication system into seven layers, each performing specific tasks toward communicating with other systems [5], [1]. The OSI is a reference model and is maintained by the ISO under the identity of ISO/IEC 7498-1.

- (i) **Physical Layer:** This is a media layer and is also referred to as layer 1 of the OSI model. The physical layer is responsible for taking care of the electrical and mechanical operations of the host at the actual physical level. These operations include or deal with issues relating to signal generation, signal transfer, voltages, the layout of cables, physical port layout, line impedances, and signal loss. This layer is responsible for the topological layout of the network (star, mesh, bus, or ring), communication mode (simplex, duplex, full duplex), and bit rate control operations. The protocol data unit associated with this layer is referred to as a *symbol*.

- (ii) **Data Link Layer:** This is a media layer and layer 2 of the OSI model. The data link layer is mainly concerned with the establishment and termination of the connection between two hosts, and the detection and correction of errors during communication between two or more connected hosts. IEEE 802 divides the OSI layer 2 further into two sub-layers [2]: Medium access control (MAC) and logical link control (LLC). MAC is responsible for access control and permissions for connecting networked devices; whereas LLC is mainly tasked with error checking, flow control, and frame synchronization. The protocol data unit associated with this layer is referred to as a *frame*.
- (iii) **Network Layer:** This layer is a media layer and layer 3 of the OSI model. It provides a means of routing data to various hosts connected to different networks through logical paths called virtual circuits. These logical paths may pass through other intermediate hosts (nodes) before reaching the actual destination host. The primary tasks of this layer include addressing, sequencing of packets, congestion control, error handling, and Internetworking. The protocol data unit associated with this layer is referred to as a *packet*.
- (iv) **Transport Layer:** This is layer 4 of the OSI model and is a host layer. The transport layer is tasked with end-to-end error recovery and flow control to achieve a transparent transfer of data between hosts. This layer is responsible for keeping track of acknowledgments during variable-length data transfer between hosts. In case of loss of data, or when no acknowledgment is received, the transport layer ensures that the particular erroneous data segment is re-sent to the receiving host. The protocol data unit associated with this layer is referred to as a *segment* or *datagram*.
- (v) **Session Layer:** This is the OSI model's layer 5 and is a host layer. It is responsible for establishing, controlling, and terminating of communication between networked hosts. The session layer sees full utilization during operations such as remote procedure calls and remote sessions. The protocol data unit associated with this layer is referred to as *data*.
- (vi) **Presentation Layer:** This layer is a host layer and layer 6 of the OSI model. It is mainly responsible for data format conversions and encryption tasks such that the syntactic compatibility of the data is maintained across the network, for which it is also referred to as the *syntax layer*. The protocol data unit associated with this layer is referred to as *data*.
- (vii) **Application Layer:** This is layer 6 of the OSI model and is a host layer. It is directly accessible by an end-user through software APIs (application program interfaces) and terminals. Applications such as file transfers, FTP (file transfer protocol), e-mails, and other such operations are initiated from this layer. The application layer deals with user authentication, identification of communication hosts, quality of service, and privacy. The protocol data unit associated with this layer is referred to as *data*.

A networked communication between two hosts following the OSI model is shown in Figure 1.3. Table 1.2 summarizes the OSI layers and their features, where PDU stands for protocol data unit.

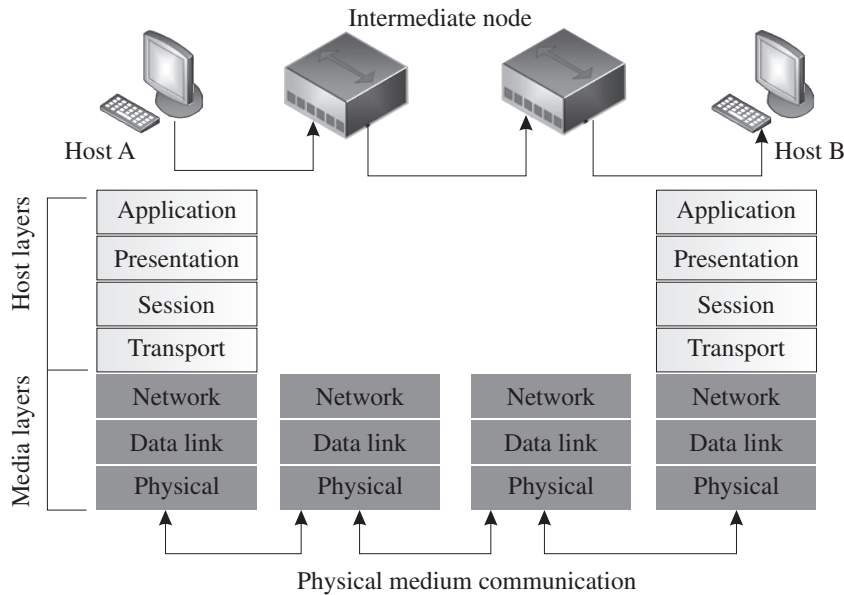


Figure 1.3 Networked communication between two hosts following the OSI model

Check yourself

Ethernet, FDDI, B8ZS, V.35, V.24, RJ45, PPP, FDDI, ATM, IEEE 802.5/ 802.2, IEEE 802.3/802.2, HDLC, Frame Relay, AppleTalk DDP, IP, IPX, NFS, NetBios names, RPC, SQL, ASCII, EBCDIC, TIFF, GIF, PICT, JPEG, MPEG, MIDI, NFS, SNMP, Telnet, HTTP, FTP

Table 1.2 Summary of the OSI layers and their features

Layer	Name	Location	PDU	Function	Examples
1	Physical	Media	Symbol	Communication over physical medium	Ethernet, FDDI, B8ZS, V.35, V.24, RJ45
2	Data link	Media	Frame	Reliability of communication over physical medium	IEEE 802.5 / 802.2, IEEE 802.3 / 802.2, PPP, HDLC, Frame Relay, ATM, FDDI
3	Network	Media	Packet	Structuring of data and routing between multiple nodes	DDP, IP, AppleTalk, IPX
4	Transport	Host	Segment	Reliability of communication over networks or between hosts	SPX, TCP, UDP
5	Session	Host	Data	Establishment, management, and termination of remote sessions	NetBios names, NFS, RPC, SQL
6	Presentation	Host	Data	Syntactic conversion of data and encryption	Encryption, ASCII, MIDI, PICT, JPEG, EBCDIC, TIFF, GIF, MPEG
7	Application	Host	Data	User identification, authentication, privacy, and quality of service	SNMP, Telnet, WWW browsers, HTTP, NFS, FTP

1.3.2 Internet protocol suite

The Internet protocol suite is yet another conceptual framework that provides levels of abstraction for ease of understanding and development of communication and networked systems on the Internet. However, the Internet protocol suite predates the OSI model and provides only four levels of abstraction: 1) Link layer, 2) Internet layer, 3) transport layer, and 4) application layer. This collection of protocols is commonly referred to as the TCP/IP protocol suite as the foundation technologies of this suite are transmission control protocol (TCP) and Internet protocol (IP) [3], [4], [6]. The TCP/IP protocol suite comprises the following four layers:

Points to ponder

The development of the TCP/IP protocol suite is originally attributed to DARPA, which is part of the United States Department of Defence. The Internet protocol suite or the TCP/IP protocol suite is sometimes also referred to as the Department of Defence (DoD) model.

- (i) **Link Layer:** The first and base layer of the TCP/IP protocol suite is also known as the network interface layer. This layer is synonymous with the collective physical and data link layer of the OSI model. It enables the transmission of TCP/IP packets over the physical medium. According to its design principles, the link layer is independent of the medium in use, frame format, and network access, enabling it to be used with a wide range of technologies such as the Ethernet, wireless LAN, and the asynchronous transfer mode (ATM).
- (ii) **Internet Layer:** Layer 2 of the TCP/IP protocol suite is somewhat synonymous to the network layer of the OSI model. It is responsible for addressing, address translation, data packaging, data disassembly and assembly, routing, and packet delivery tracking operations. Some core protocols associated with this layer are address resolution protocol (ARP), Internet protocol (IP), Internet control message protocol (ICMP), and Internet group management protocol (IGMP). Traditionally, this layer was built upon IPv4, which is gradually shifting to IPv6, enabling the accommodation of a much more significant number of addresses and security measures.
- (iii) **Transport Layer:** Layer 3 of the TCP/IP protocol suite is functionally synonymous with the transport layer of the OSI model. This layer is tasked with the functions of error control, flow control, congestion control, segmentation, and addressing in an end-to-end manner; it is also independent of the underlying network. Transmission control protocol (TCP) and user datagram protocol (UDP) are the core protocols upon which this layer is built, which in turn enables it to have the choice of providing connection-oriented or connectionless services between two or more hosts or networked devices.

- (iv) **Application Layer:** The functionalities of the application layer, layer 4, of the TCP/IP protocol suite are synonymous with the collective functionalities of the OSI model's session, presentation, and application layers. This layer enables an end-user to access the services of the underlying layers and defines the protocols for the transfer of data. Hypertext transfer protocol (HTTP), file transfer protocol (FTP), simple mail transfer protocol (SMTP), domain name system (DNS), routing information protocol (RIP), and simple network management protocol (SNMP) are some of the core protocols associated with this layer.

A networked communication between two hosts following the TCP/IP model is shown in Figure 1.4

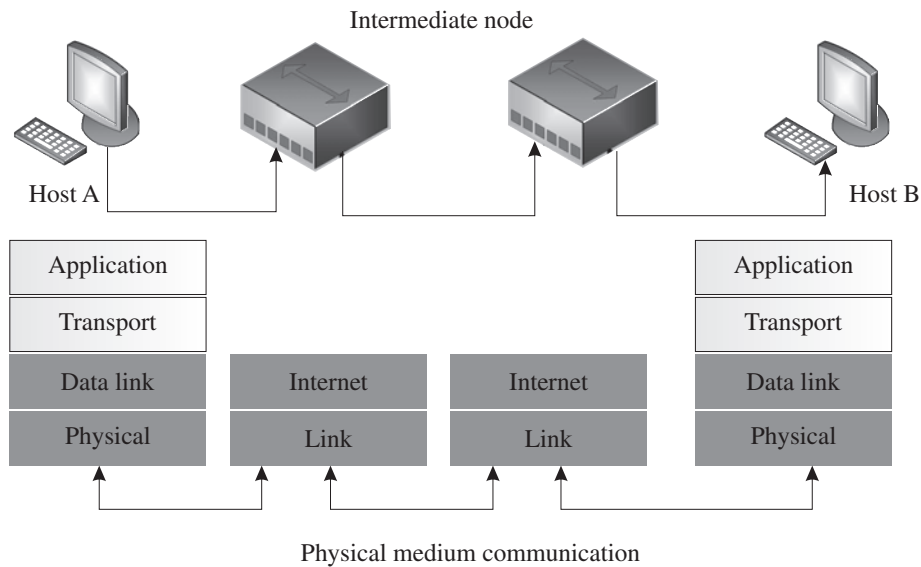


Figure 1.4 Networked communication between two hosts following the TCP/IP suite

1.4 Addressing

Addressing in networked devices plays a crucial role in ensuring the delivery of packets to the designated/intended receivers. The addressing scheme is synonymous with postal addresses used in real-life scenarios. Addressing mechanisms can be divided into two parts: one focusing on data link layer addressing, while the other focuses on network layer addressing.

1.4.1 Data link layer addressing

Data link layer addressing deals with media access control (MAC) addresses of devices, which work at the MAC sub-layer of the data link layer.